

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Award	Centre Number	Candidate Number	
Wednesday 26 June 2019			
Morning (Time: 3 hours)		Paper Reference 9811/01	
Advanced Extension Award Mathematics			
You must have: Mathematical Formulae and Statistical Tables			Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- **Calculators may not be used.**
- You must **show all your working.**
- Answers should be given in as simple a form as possible. e.g. $\frac{2\pi}{3}$, $\sqrt{2}$, $3\sqrt{2}$.

Information

- The total mark for this paper is 100 of which **7** marks are for style and clarity of presentation.
- The style and clarity of presentation marks will be indicated as **(+S1) or (+S2)**.
- There are 7 questions in this question paper.
- The marks for each question are shown in brackets.
- The total mark for each question is shown at the end of the question.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. (a) By writing $u = \log_4 r$, where $r > 0$, show that

$$\log_4 r = \frac{1}{2} \log_2 r \quad (2)$$

- (b) Solve the equation

$$\log_4 (5x^2 - 11) = \log_2 (3x - 5) \quad (5)$$



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. The discrete random variable X follows the binomial distribution

$$X \sim \text{B}(n, p)$$

where $0 < p < 1$. The mode of X is m .

- (a) Write down, in terms of m , n and p , an expression for $P(X = m)$ (1)
- (b) Determine, in terms of n and p , an interval of width 1, in which m lies. (5)
- (c) Find a value of n where $n > 100$, and a value of p where $p < 0.2$, for which X has **two** modes.
For your chosen values of n and p , state these two modes. (2)



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Question 2 continued

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(Total for Question 2 is 8 marks)



3. Given that $\phi = \frac{1}{2}(\sqrt{5} + 1)$,

(a) show that

(i) $\phi^2 = \phi + 1$

(ii) $\frac{1}{\phi} = \phi - 1$

(4)

(b) The equations of two curves are

$$y = \frac{1}{x} \quad x > 0$$

$$\text{and} \quad y = \ln x - x + k \quad x > 0$$

where k is a positive constant.

The curves touch at the point P .

Find in terms of ϕ

(i) the coordinates of P ,

(ii) the value of k .

(6)

(+S1)



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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 11 marks)



4. (a) Prove the identity

$$(\sin x + \cos y) \cos(x - y) \equiv (1 + \sin(x - y))(\cos x + \sin y) \quad (5)$$

(b) Hence, or otherwise, show that

$$\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta} \equiv \frac{1 + \tan \theta}{1 - \tan \theta} \quad (6)$$

(c) Given that $k > 1$, show that the equation $\frac{\sin 5\theta + \cos 3\theta}{\cos 5\theta + \sin 3\theta} = k$ has a unique solution in the interval $0 < \theta < \frac{\pi}{4}$ (4)

(+S2)



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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 17 marks)



5. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively, relative to an origin O , and are such that OAB is a triangle with $OA = a$ and $OB = b$.

The point C , with position vector $\mathbf{c}$, lies on the line through O that bisects the angle AOB .

- (a) Prove that the vector $b\mathbf{a} - a\mathbf{b}$ is perpendicular to $\mathbf{c}$. (4)

The point D , with position vector $\mathbf{d}$, lies on the line AB between A and B .

- (b) Explain why $\mathbf{d}$ can be expressed in the form $\mathbf{d} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$ for some scalar λ with $0 < \lambda < 1$ (2)

- (c) Given that D is also on the line OC , find an expression for λ in terms of a and b only and hence show that

$$DA : DB = OA : OB \quad (8)$$

(+S2)



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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 16 marks)



6. Figure 1 shows a sketch of part of the curve with equation $y = x \sin(\ln x)$, $x \geq 1$

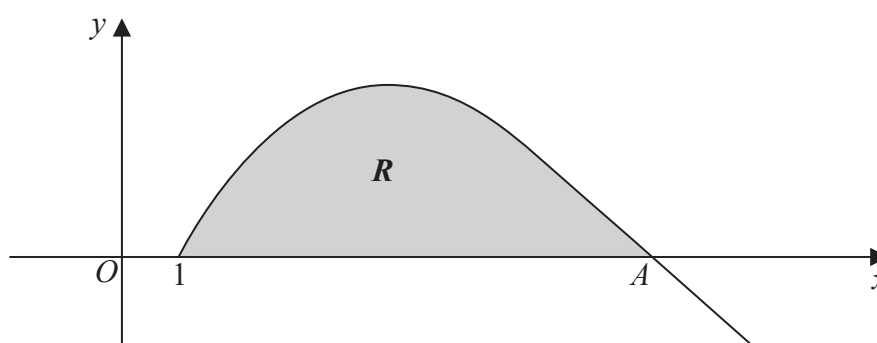


Figure 1

For $x > 1$, the curve first crosses the x -axis at the point A .

- (a) Find the x coordinate of A . (3)

- (b) Differentiate $x \sin(\ln x)$ and $x \cos(\ln x)$ with respect to x and hence find

$$\int \sin(\ln x) dx \text{ and } \int \cos(\ln x) dx \quad (7)$$

- (c) (i) Find $\int x \sin(\ln x) dx$.

- (ii) Hence show that the area of the shaded region R , bounded by the curve and the x -axis between the points $(1, 0)$ and A , is

$$\frac{1}{5}(e^{2\pi} + 1) \quad (9)$$



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 19 marks)



7. Figure 2 shows a rectangular section of marshland, $OABC$, which is a metres long by b metres wide, where $a > b$.

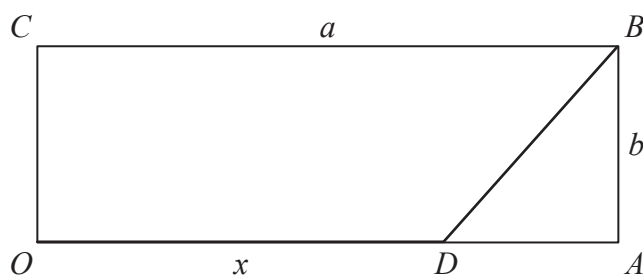


Figure 2

Edgar intends to get from O to B in the shortest possible time. In order to do this, he runs along edge OA for a distance x metres ($0 \leq x < a$) to the point D before wading through the marsh directly from D to B .

Edgar can wade through the marsh at a constant speed of 1 ms^{-1} , and he can run along the edge of the marsh at a constant speed of $\lambda \text{ ms}^{-1}$, where $\lambda > 1$

- (a) By finding an expression in terms of x for the time taken, t seconds, for Edgar to reach B from O , show that

$$\frac{dt}{dx} = \frac{1}{\lambda} - \frac{a-x}{\sqrt{(a-x)^2 + b^2}} \quad (5)$$

- (b) (i) Find, in terms of a , b and λ , the value of x for which $\frac{dt}{dx} = 0$

(ii) Show that this value of x lies in the interval $0 \leq x < a$ provided $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

- (iii) For λ in this range, show that the value of x found in (b)(i) gives a minimum value of t . (8)

- (c) Find the minimum time taken for Edgar to get from O to B if

(i) $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

(ii) $1 < \lambda < \sqrt{1 + \frac{b^2}{a^2}}$ (4)

Edgar's friend, Frankie, also runs at a constant speed of $\lambda \text{ ms}^{-1}$. Frankie runs along the edges

OA and AB . Given that $\lambda \geq \sqrt{1 + \frac{b^2}{a^2}}$

- (d) find the range of values of λ for which Frankie gets to B from O in a shorter time than Edgar's minimum time. (3)

(+S2)



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Question 7 continued

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Question 7 continued

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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 22 marks)

FOR STYLE, CLARITY AND PRESENTATION: 7 MARKS
TOTAL FOR PAPER: 100 MARKS



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